PREPARED FOR AURUMIX

Aurumix Gold Savings Simulation

How the simulation is built. The customer, the score, the gold, the float, and what happens each month.

tokenomics.net

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Contents

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What the simulation is for

What the simulation is for, and why it asks what has to be true rather than what will happen.

The business in one paragraph

Aurumix sells gold by the gram. A customer commits to a monthly amount from USD 20. Every dollar buys allocated physical gold in a Dubai vault. One AURX token is one gram.

The customer pays a 5% entry fee and nothing else. Behind the savings product sit a card, a credit line against the gold, a family plan, and a platform fee from B2B partners who white-label the product.

What the revenue model already says

The Phase 4 revenue model is a spreadsheet. It produces one seven-year projection from about ninety assumptions. There is no customer data yet. No waitlist, no pilot, no partner in the pipeline.

The two questions

The simulation does not ask what will happen. It asks two harder questions, and the results document is structured around them.

- What has to be true for this business to make money?
- What happens when something goes badly wrong, and can more money fix it?

Everything else follows from these. How much to raise, how much can be given back to loyal customers, and which changes to the plan create the most profit are all answers to the first question, not questions of their own.

Why a threshold and not a forecast

A forecast built on unsourced inputs is a guess with decimal places. A threshold is different. It says "you need this many paying customers," and that number does not depend on whether the market funnel is right.

That is the only honest deliverable at this stage. The document is built around it.

02

One system, run whole

A line-for-line port of the revenue model, and a population of individual customers that grows the same book.

The principle

The revenue model is a description of the business. The simulation is the business running.

Every rule in the revenue model lives inside the simulation, executing on simulated customers, month by month. The vault bill is charged on the grams they actually hold. Card fees are earned on what they actually spend. The loyalty discount is priced at the tier each of them actually reached. Nothing is computed as an average customer times a head count.

We call this the twin. One system, one engine.

What the twin takes from the revenue model

Parameters. Prices, fees, the licence costs, the vault contract, the card economics, the marketing budget, the salary lines. These are facts about the business. Keeping a second copy of them would only create two versions of the truth.

What the twin computes for itself

Everything behavioural. Who arrives, who pays, who reaches a tier, who leaves, who refers, what each of them costs and earns. The spreadsheet approximates these with one average customer, one flat churn rate, and one blended discount. The twin replaces each approximation with the individual behaviour underneath it, and the rest of this document describes exactly how.

03

The frame

Region, ticket, payment archetype, payment rail and entry door. What each customer carries at birth.

Method	agent-based simulation inside a Monte Carlo
Horizon	84 months, seven years
Month 1	January 2027
Step	one month, every month, for everything
Paths	2,000, each a full rerun of the population
Customers per path	not fixed: an outcome of each path, around 170,000 at the median
Resolution	one simulated agent per customer in the final run; 1 agent per 10 while iterating
Drawn inputs per path	75 parameters, plus a gold path and a partner history
Randomness	fixed seeds, so every run is exactly reproducible

Agent-based means the unit of computation is one customer, carried through their whole history. Monte Carlo means the entire seven years is rerun 2,000 times, each time under different assumptions drawn from stated ranges. Section 10 covers the drawing. Everything before it covers what happens inside one path.

04

The customer

Six consecutive payments open the gate. The score is computed from three facts on the ledger, not looked up.

Each simulated customer is created with five properties: a region, a monthly ticket, a payment personality, a payment rail, and an entry door. Their whole later life follows from these plus luck.

Region

One of the UAE, Oman and Bahrain together, or India. Region sets the average ticket, the acquisition cost, and when the market opens. India and the UAE open in month one. Oman and Bahrain open in month thirteen.

Ticket: how much they mean to save

The revenue model uses one number per region, USD 33.60 in the UAE. Savings books are not like that: most of the book saves little and a thin tail saves a lot. So each customer draws a personal ticket from a curve fitted to two facts, the regional mean and the share at the floor:

$$\text{ticket} = \max\left(20,\ X\right), \qquad X \sim \text{Lognormal}(\mu, \sigma)$$

Where:

- μ and σ are solved numerically so that $P(X \leq 20) = 0.30$ and $E[\max(20, X)]$ equals the regional mean
- 20 is the USD floor, below which a payment is refused

Each region fits its own curve to its own mean; the USD 20 floor and the 30% floor share are common to all three. The two facts lock the curve. If 30 customers in 100 pay exactly USD 20, that is USD 600 of a USD 3,360 total, so the other 70 must average USD 39.43. For the UAE this solves to $\sigma = 0.576$, and the top tenth of customers ends up contributing about a quarter of all money saved.

The amount actually declared in a given month wobbles around the personal ticket:

$$\text{amount}_t = \max\left(20,\ \text{ticket} \times e^{\,0.25\,\varepsilon_t}\right), \qquad \varepsilon_t \sim \mathcal{N}(0, 1)$$

A saver at the floor pays exactly the floor. Whether they pay at all is the personality's job, below.

Payment personality: whether they pay, and whether they stay

Five types. Each type carries a monthly probability of paying and its own monthly hazard of stopping the plan:

ARCHETYPE	SHARE	PAYS IN A GIVEN MONTH	OWN MONTHLY LAPSE HAZARD
Perfect payer	10%	99.5%	0.00%
Occasional misser	35%	93%	0.46%
Alternating misser	12%	55%	1.19%
Reducer	13%	97%	0.13%
Early lapser	30%	60%	13.25%

Each month, every active customer:

- **pays** with probability $p_t = \max(p_0 \cdot d^{\text{age}},\ p_{\min})$, where p_0 is the type's rate from the table and age their months on the platform. d is a fade dial and $p_{\min}$ its floor. Both are off at base ($d = 1, p_{\min} = 0$), so the base case pays at the flat rate; the dials exist for sweeps that test whether enthusiasm wears off
- **leaves** with probability equal to their own hazard plus a background hazard of 1.06% per month that applies to everyone

Together the five types imply the book pays in roughly 78% of the months it could. That single property separates this model from the spreadsheet, whose paying customers pay every month by construction.

Where the five types come from

They are not observed segments. They are a mixture tuned to reproduce one target curve. A cohort should have 63% still paying at month 13, then 49, 41, 34 and 29% at months 25, 37, 49 and 61. When the Monte Carlo draws a different month-13 persistency, every hazard is rescaled by a single factor so the cohort hits it. One factor, not five free parameters, so the mix and its ordering survive.

Why the curve keeps falling

The 63% level is anchored to Indian life insurance persistency, published by the regulator at the same checkpoints. But insurance curves flatten after year one, and ours does not, on purpose. An insurance saver who quits forfeits surrender value, so the penalty does the retaining. An Aurumix saver forfeits nothing: the gold is theirs and leaving is free. Easy exit means we should expect weaker long-run retention than a penalty-locked product, so the twin borrows the anchor's level and rejects its flattening. This is a deliberately conservative choice, and persistency is drawn per path in any case.

Rail: how the money moves

Each customer is assigned a rail at birth: a standing instruction ("set and forget"), or a manual payment each month. At base, 30% of joiners take the standing instruction; the share is swept, and it is a lever Aurumix controls.

The rail changes who the customer is, not just how they pay. A customer on the standing instruction has a 35% chance of being re-drawn from the disciplined types (perfect payer, occasional misser, reducer). That is the modelled link between removing friction and clearing six consecutive payments. Both numbers are unsourced and swept.

Door

How they arrived: marketing, an agent, or a referral. The door carries the acquisition cost attached to that customer and drives the referral economics in section 7.

Spot purchases: buying outside the plan

On top of the monthly plan, an active saver can make one-off purchases. Each month, each saver has a chance of one:

$$P(\text{spot this month}) = \text{attach}_r \times m_a \times \frac{\text{freq}}{12}, \qquad \text{amount} = \text{ticket}_r \times m_t \times \text{noise}$$

Where:

- attach_r is the regional share of savers who spot-buy at all: 12% UAE, 10% Oman and Bahrain, 35% India
- freq is how often a spot buyer buys, 1.7 times a year at base
- ticket_r is the regional spot amount: USD 190, 145 and 40
- m_a and m_t are drawn multipliers on attach and amount, and freq is drawn too, so spot behaviour varies per world
- noise spreads individual purchases around the regional average, the same way monthly tickets wobble

A spot purchase pays the entry fee at the customer's tier and becomes gold like any other money in. One simplification to know: spot rides on the existing saver book, per the revenue model's attach structure. A spot-only customer who never opens a plan is not modelled, because no number exists yet for how many there would be.

What a customer does with their gold

Two exits exist for gold, and neither involves physical delivery, because Aurumix does not offer physical delivery.

- **Redemption**: the customer sells gold back for cash. Each month each customer has a 0.5% chance of redeeming (a 6% annual rate spread across twelve months), and a redemption sells a quarter of their holding. Lapsed customers who still hold gold redeem at 1.6 times that rate, 0.8% a month. Redeemed metal returns to Aurumix and is recycled into the float.
- **Self-custody**: the customer moves AURX tokens to their own wallet. Also a 0.5% monthly chance, also a quarter of the holding. Nothing is redeemed: the gold stays in the vault, the vault bill keeps running on it, and it still counts in AUM. What changes is the account. The ICS score and the credit line are tied to the account, so tokens moved out stop counting toward the score and stop backing the credit line.

Both exits count against the loyalty score, because the score is tied to the account. Gold sold back is gone. Tokens moved to your own wallet no longer count toward it. Section 6 shows how.

05

A month in the system

Five benefits priced by tier, generated from three dials, with one rule only an agent model can enforce.

The core loop. Every month, in this order.

1. **Gold moves** to this month's price.
2. **New customers arrive**, channel by channel, against the market ceiling (section 7).
3. **Each new customer** is created with region, ticket, personality, rail and door.
4. **Existing customers decide whether to pay**, at their personality's odds.
5. **Monthly payments and any spot purchases become gold** at this month's price, minus the entry fee at that customer's tier.
6. **Some customers redeem for cash, some move tokens to their own wallet**, per the rates above.
7. **Streaks update.** Six consecutive payments opens the loyalty gate, once, permanently.
8. **The loyalty score recalculates** for everyone (section 6).
9. **Tiers reassign** from the score.
10. **Cards, family plans and credit lines** run for whoever holds them, each against that customer's own gold.
11. **Some customers leave**, at their own hazard plus the background rate.
12. **Partners sign**, or fail to (section 8).
13. **The ledger closes.** Five revenue streams plus the partner fee. Against them: the fabrication premium on net new metal, the vault bill on all metal held, licences, acquisition, the card programme, and contingency on the uncertain costs. What remains is profit, and the cash line follows it.

06

The loyalty ladder

Stopping the SIP, letting the card go dormant, repaying credit and redeeming gold each tick on their own clock.

The gate

Six consecutive monthly payments. One property of one person's history, which is the fact that forces a simulation of individuals: six-in-a-row cannot be recovered from an average. The gate opens once and never closes.

The score

Once gated, a customer's Integrated Continuity Score is recomputed every month from three ingredients:

$$\text{Score} = \min(\text{Record},\ \text{Standing}) \times \text{Retention}$$

Where:

- **Record** rewards accumulated history: it climbs by $\frac{50}{12}$ per counted month to 50 at one year, then by $\frac{50}{48}$ per month to 100 at five years. Five years of saving is a complete record.
- **Standing** rewards the recent twelve months: the count of paid months in the last twelve, times $\frac{100}{12}$.
- **Retention** penalises gold leaving the account, whether sold back or moved to the customer's own wallet. Up to 30% in a year and nothing happens; beyond that, $\text{Retention} = 1 - \frac{\text{sold}-0.30}{0.70}$.
- Once gated, the score never falls below 25.

The $\min$ means neither a long history nor a good recent year can substitute for the other. Tiers follow fixed cutoffs: Silver at 25, Gold at 50, Platinum at 75, Sovereign at 100. Sovereign therefore requires a five-year record and a perfect recent year at once, which makes it nearly unreachable inside the horizon by construction: expensive to earn, cheap to offer.

What a tier buys

	UNTIERED	SILVER	GOLD	PLATINUM	SOVEREIGN
Entry fee	5.0%	4.5%	4.0%	3.5%	3.0%
Family discount	0%	10%	20%	35%	50%
FX margin	2.0%	2.0%	1.5%	1.25%	1.0%
Borrow against gold	50%	50%	65%	72.5%	80%
Card rewards	0%	0.15%	0.45%	0.60%	0.75%

The 5% entry fee is a hard cap. Nobody ever pays more; the ladder only gives back. The ladder itself is a swept design object: its ceiling, its steepness and its breadth are all dials the simulation prices.

How the giveback is charged

Every component is priced per customer per month, as activity times the gap between the standard price and that customer's tier price:

- entry discount: contributions × (5.0% − tier entry fee)
- FX discount: foreign card spend × (2.0% − tier FX margin)
- rewards: card spend × tier rewards rate
- family discount: family fee × tier discount

The cashback cap

One rule applies to the gold cashback, and to the cashback only: it is capped against the customer's own card transactions. Cashback on a transaction never exceeds what that transaction earns Aurumix, which is the interchange net of the scheme's share plus the FX margin.

At quoted rates the structure satisfies this by itself. A dollar of spend earns Aurumix 1.1 to 1.4 cents and the top cashback tier pays 0.75 cents, so the cap never binds in the base case. It sits in the simulation as a safety rail. When a sweep makes the ladder more generous, or a path draws interchange low, this is what stops the model paying customers to spend.

07

How customers arrive

Gold as geometric Brownian motion, partners as lumpy arrivals, and every other input drawn from the client's own scenario table.

Three channels and a brake decide each region's new customers each month:

$$\text{new}_t = \Big(\underbrace{\tfrac{B_t}{\text{CAC}_t}(1+o)}_{\text{marketing}} + \underbrace{W_t \cdot \tfrac{\rho}{12} \cdot v}_{\text{referrals}} + \underbrace{a_t \cdot g \cdot r_t}_{\text{agents}}\Big) \times \underbrace{\Big(1 - \tfrac{\text{acquired so far}}{\text{ceiling} \times m}\Big)}_{\text{saturation}} \times \text{season}_t$$

Marketing

B_t is the region's slice of the monthly marketing budget and o an organic uplift of 25%. CAC_t, the cost of reaching one customer, has two parts:

$$\text{CAC}_t = \underbrace{\Big[c_1 + (c_7 - c_1)\tfrac{y-1}{6}\Big]}_{\text{planned ramp}} \times \underbrace{\Big[1 + \kappa\Big(\tfrac{B_t}{60{,}000}\Big)^{0.7}\Big]}_{\text{channel exhaustion}}$$

The ramp is the plan: UAE cost falls from USD 85 to USD 55 by year seven as the brand matures. The second term pushes back: spend rises eighteen-fold over the horizon, cheap channels exhaust, and each further customer costs more to reach. κ is 0.35 at base and is drawn between 0.15 and 0.60, so the raise prices not knowing how strong the exhaustion is.

Referrals

W_t is the book's referral capacity: every paying customer's individual propensity, summed. Propensity is zero for the first three months, then ramps to full by month twelve. It rises up the tier ladder too, from 1.0 untiered to 1.75 at the top. ρ is the referral rate, 0.6 per fully established customer per year, and v the 62% conversion of referrals into customers.

So a young book refers at about a third of the rate its head count suggests; a matured book at the full rate. Referrals also cost money: the reward is 30% of the referee's first six months of entry fees.

Agents

a_t field agents (the plan runs them mostly in India), each converting $g = 6$ customers a month, times a productivity ramp r_t that starts at 60% in year one.

Saturation and the calendar

Each region has an addressable ceiling, an estimate of how many people can ever be reached, built from population filter assumptions. Acquisition scales down linearly as the region fills. A number that binding does not get to be a point estimate. The multiplier m is drawn between 0.6 and 1.55 per path, so every result already prices the market being two-thirds or one-and-a-half times the estimate. Finally the calendar. A twelve-month seasonality shape follows the year's savings and spending events, normalised so a year sums to twelve, with a 10% random wobble on each month.

08

Partners

The order of operations inside every one of the eighty-four months.

What a partner is worth

A B2B partner white-labels the product. Its yearly value is four numbers multiplied:

$$V = U \times a \times h \times f$$

Where:

- U is the partner's user base, 900,000 at base
- a is the share of those users who adopt the product, 6%
- h is the average gold value an adopter holds, USD 350
- f is the platform fee on the total, 0.75% a year

At base: $900{,}000 \times 0.06 \times 350 \times 0.0075 =$ **USD 141,750 per partner per year**. All four numbers carry Aggressive and Conservative ranges in the revenue model and all four are drawn per path. On top of the world's draw, each partner that signs gets a personal size factor averaging one, applied to its gold under management. So a book can hold one whale and three minnows around the same average.

Whether a big partner is big through more users or keener ones is not separated. The fee only ever sees the product of the two.

A signed partner does not arrive at full power: its users adopt over time. Revenue climbs linearly to full over a ramp of about 18 months, drawn between 12 and 24 per path. The monthly stream is then each signed partner's gold under management, scaled by how far through its ramp it is:

$$s_{6,t} = \sum_{i \text{ signed}} \mathrm{AUM}_i \times \min\left(1,\ \tfrac{t - t_i + 1}{\mathrm{ramp}}\right) \times \frac{f}{12}$$

where t_i is partner i's signing month. The ramp matters more than it looks: the funding trough falls exactly when the first partners are mid-ramp, so assuming instant adoption would understate the raise.

How partners arrive

The plan says eleven partners by year seven: cumulative targets of 0, 1, 3, 5, 7, 9, 11 by year end. In the twin they arrive as discrete signings, not a straight line. For each year y with Δ_y planned additions:

$$\mathrm{signed}_y = \begin{cases} 0 & \text{with probability } 0.25 \quad \text{(a dead year)} \\ \mathrm{Poisson}(\Delta_y / 0.75) & \text{with probability } 0.75 \end{cases}$$

The Poisson mean is grossed up by the dead-year odds, so the expected signings equal the plan: $\mathbb{E}[\mathbf{signed}_y] = \Delta_y$. Dead years and clusters widen the range of outcomes without quietly moving the middle. Doubting the plan itself would be an assumption, and assumptions get declared, not hidden inside noise.

Once signed, a partner's fee runs every month from its signing month.

09

Gold and the float

The float sized from the payment calendar with dealer lead time, service level, carry and mark-to-market. Unhedged.

The price

One gold price for the whole system, following geometric Brownian motion:

$$S_{t+1} = S_t \exp\left[\left(\mu - \tfrac{1}{2}\sigma^2\right)\Delta t + \sigma\sqrt{\Delta t}\,\varepsilon_t\right], \qquad \varepsilon_t \sim \mathcal{N}(0,1)$$

Where:

- μ is the drift, 8.1% a year, the appreciation assumption carried through from the revenue model
- σ is volatility, 15% a year, the long-run realised figure for gold in USD, swept 10 to 22%
- Δt is one month

Every part of the system reads this same path: purchases, the vault bill, collateral values, credit limits, the float.

The float

Aurumix keeps a working inventory of gold so purchases settle instantly while replacement bars are in transit. Inside each path the requirement is one bar plus enough to cover a buffer of days at that month's actual purchase rate, with a two-bar minimum.

A separate daily study prices the policy as a standard inventory problem. Order up to a level S that covers expected demand over the delivery lead time, plus a safety margin:

$$S = \mathbb{E}[\text{demand over lead time}] + z \cdot \sigma_d\sqrt{L} + \text{bar},$$

rounded up to whole bars, where z sets the service level and L is the lead time. It runs on the twin's own monthly purchase series, including paydays and seasonality, and prices carry cost and unhedged mark-to-market on the same gold paths.

10

Two thousand worlds

Margin calls tracked by vintage, and why the loan-to-value rung decides everything.

Each path is one version of the next seven years. What varies between paths, and how.

Seventy-five drawn parameters

Every input carrying an Aggressive and a Conservative value in the Phase 4 revenue model is drawn. The distribution is Beta-PERT: Base is the most likely value, and Aggressive and Conservative bracket the range,

$$x \sim \mathrm{lo} + \mathrm{Beta}(\alpha, \beta) \times (\mathrm{hi} - \mathrm{lo}), \qquad \alpha = 1 + 4\frac{\mathrm{mode}-\mathrm{lo}}{\mathrm{hi}-\mathrm{lo}}, \quad \beta = 1 + 4\frac{\mathrm{hi}-\mathrm{mode}}{\mathrm{hi}-\mathrm{lo}}$$

Those bands come from the Phase 4 revenue modelling. Researched where a source existed, set by judgement where none did. They are our own estimates, not figures supplied by anyone, and the document says so wherever one is load-bearing. Reusing them here keeps the simulation and the revenue model arguing from the same evidence rather than from two sets of numbers. The rule is opt-out, not opt-in: any input with a range is drawn, so the raise number carries it. Eleven parameters are excluded, each with a written reason in the code, in three groups:

- **replaced by structure**: the partner count (a discrete arrival process now), the gold band (the price process carries it)
- **derived**: quantities recomputed from their drawn components each path, so they move with their inputs
- **superseded by behaviour**: the workbook's blended loyalty rates and "share who ever tier". The twin prices each customer at their own tier, so a blended rate has nothing left to describe.

Three Phase 5 parameters have no workbook row, so they carry their own declared bands. The market ceiling multiplier, 0.6 to 1.55. The channel-exhaustion strength, 0.15 to 0.60. The partner adoption ramp, 12 to 24 months.

And per path

A fresh gold path, a fresh partner history, a fresh demand wobble, and a full rerun of the population: who arrives, who pays, who gates, who refers, who leaves. Nothing is reused between paths except the fixed seeds that make every result reproducible.

11

What comes out

The steady-state threshold, with retail customers and B2B partners kept apart.

Stack 2,000 paths and read the spread.

OUTPUT	MEANING
Safe raise	the funding trough deep enough to survive 9 paths in 10
Break-even odds	share of paths whose cumulative profit turns positive by year 7
Profit spread	year-seven net profit at the 10th, 50th and 90th percentile
Book size	paying customers at month 84

The funding line itself is monthly: cumulative losses plus the capital tied up in the float, card prefunding and regulatory capital, with the raise sized off its deepest point. The results document carries the numbers. This document stops at how they are produced.

12

What it takes to be profitable

Seven scenarios, including a redemption run built as a jump rather than a rate.

The profitability question is a threshold, and it is computed at steady state, where the book is flat and acquisition only replaces the customers who leave.

$$N^* = \frac{F_c + kF_u}{r - (v_c + k\,v_u)} \qquad\qquad K^* = \frac{F_c + kF_u}{A \cdot f}$$

Where:

- N^* is paying customers needed for retail to stand alone, K^* is B2B partners needed to cover the fixed base by themselves
- r is retail revenue per paying customer per year, taken from the twin's own mature book, so it already reflects missed months
- v_c and v_u are the per-customer costs, split by whether they can surprise us
- F_c and F_u are the yearly fixed costs, split the same way
- A is assets under management per partner, f the platform fee
- k is the contingency multiplier, reported at 1.15, 1.30 and 1.50

Per-customer costs include each customer's share of replacing the ones who leave: annual churn, measured from the twin's own mature book, times the cost of winning a replacement.

Why contingency lands on only half the costs

Contingency buffers costs nobody has thought of. Apply it to every line and you are pricing doubt about things nobody doubts. So each cost is classified once:

	CERTAIN, NO BUFFER	UNCERTAIN, BUFFERED
Per customer	fabrication premium, vault fee, KYC checks, redemption handling, loyalty giveback, agent commission, referral rewards	marketing to replace leavers, card programme
Fixed	VARA and DMCC licences	technology build and maintenance, insurance, audits

The logic is simple. A contracted premium, a published licence fee, a vendor's price per identity check and a ladder Aurumix sets itself are known. A marketing yield, a card scheme's fees, and what technology and insurance really cost are not.

This matters more than it sounds: buffering the certain half as well was moving the retail threshold by roughly a hundred thousand customers on nothing.

It also concentrates the uncertainty where it belongs. Marketing is 92% of what it costs to win a customer, so once the certain lines are excluded, sweeping contingency is really a test of acquisition cost alone.

One reclassification came out of this. The vault contract has a daily minimum, but at this book size it does not bind. The bill is the percentage fee on metal held, and that rises with the book. So the vault is a per-customer cost, not a fixed one.

The two thresholds are reported separately and never blended, because blending them would hide whether the retail business stands alone. A threshold larger than every reachable customer in all three regions is reported as beyond the market. Printing the number instead invites someone to treat it as a target. The values, and the levers that move them, are the results document's job.

The conditions map

A single threshold answers what must be true at one set of assumptions. It does not show how the requirement moves when the assumptions do, which is the more useful question when nothing is measured.

So the same calculation is run across a grid of two variables, and only two. They are the only ones that are both decision-relevant and genuinely unknown: what a customer costs to acquire, and how many partners sign. Everything else is either a decision Aurumix makes, which is priced as a lever instead, or a figure already anchored.

Each cell is a full simulation run, not an interpolation. The axes are bounded by the Phase 4 band, not by invented extremes, and the plan's own position is marked. So it reads as a specification with a current location, not a menu of scenarios.

Per-region economics

Each region is also run on its own, with the other two switched off and its marketing budget cut to its own share. Cutting the budget matters: it keeps spend intensity identical to the blended run, so the channel-exhaustion curve bites exactly as it does in the full model. Without that, a single-region run would concentrate the whole budget in one market and invent a cost penalty the plan never incurs.

The fixed cost base is company-wide, so a region's threshold means: how many customers in that region alone would cover the whole company's fixed costs.

What to learn first, and when to change your mind

Two further layers turn the sensitivity table into something a team can act on.

What to learn first ranks each assumption by how much it moves the answer against what it costs to find out. The impact comes from the model; the cost and timing of learning are judgement, and are labelled as such in the output.

When to change your mind inverts the thresholds. For each metric observable early, the simulation is swept until the margin crosses zero, giving the level at which a decision flips rather than a description of a scenario. The sweep stays inside a range where the region still functions: pushed to extremes, marketing buys almost nobody and cost per customer divides by a vanishing denominator. Where a crossing lies outside that range, the output says so instead of extrapolating to a number. Where a metric turns out not to move the answer at all, it is reported as a non-trigger rather than given a threshold that does not bind.

13

Stress tests

Two gates that had to pass, and one published figure deliberately not chased.

Seven deliberate bad days. Each is a named scenario run through the full twin with everything else held at base, so the damage is attributable. What each one is, and how the twin models it:

SCENARIO	WHAT HAPPENS	HOW IT IS MODELLED
Gold crash	the gold price falls 30% and savers react	the GBM path steps down 30% at month 24, and the full redemption-run block below fires with it
Redemption run	a quarter of custody leaves at once	a one-month jump redeems 25% of holdings at month 24; for six months payment odds drop to 40% of normal; excess metal beyond what new purchases absorb is sold back at a 1% spread
No B2B	no partner ever signs	the partner schedule is zeroed for all seven years
Adoption failure	the product lands badly everywhere	persistency at 45%, every acquisition cost at its Conservative value, tier take-up compressed
Regulatory delay	the licence takes a year longer	revenue and growth spend shift right twelve months; the standing costs of existing (licences, insurance, audits, the technology build) keep running
Ticket compression	customers save less than planned	regional average tickets cut to USD 26.50 / 21 / 24, refitting each ticket curve against the USD 20 floor
Combined tail	run + crash + weak B2B together	the redemption run and gold crash at month 24, on a partner schedule cut roughly in half

Two modelling notes. A redemption run is a jump, not a rate: a rate-based outflow converges toward balance with inflows and can never overshoot, so it cannot represent a run. And a customer pause is a payment-odds event, not a ticket cut: tickets cannot fall below the USD 20 floor, so "saving less for a while" means skipping months.

One calibration note. The gold crash scenario includes the reaction because the price fall alone does almost nothing: customers own the gold, and revenue rides on money moving. The reaction reuses the redemption-run numbers unchanged. Nobody has measured how these savers react to a falling price, so a crash-specific calibration would be an invented number. Reuse also keeps the rows comparable: the difference between the crash and the plain run is the price fall itself.

14

How we know it is not lying

Where the evidence is strong, where it is judgement, and what is not modelled.

Reconciliation against the revenue model

The twin is deliberately not a copy of the spreadsheet, so it is not tested by matching it. It is tested by explanation: every year-seven line, twin against revenue model, with the cause of every difference written down and bounded. If a difference appears whose cause is not on record, the check fails and the work stops until it is either a documented insight or a fixed bug. Differences have exactly two legitimate sources: behaviour the spreadsheet cannot represent, and declared assumption changes.

The standing audit

Twenty-seven checks run after every change to the model. Among them: every priced range reaches the Monte Carlo. The giveback vanishes if the ladder is flattened, proving it is priced from real tiers. Drawing persistency moves the book. Peak funding is the true maximum of the monthly cash line. Every parameter the engine reads exists. Each check exists because the defect it catches is the kind that never announces itself.

Verification is not validation

All of this proves the code does what this document says. None of it proves the assumptions are right. Nothing can, until there are customers. That is why the deliverable is a threshold and a spread, not a forecast.

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